

## **Element F**

### ***Consideration of Design Viability***

A difficulty that might occur would be trying to accommodate the size of the reservoir under the structure. Since the current size of the structure can only accommodate a 30" by 60" box that would not be enough for the hydroponics unit to work at its best. We may need to reconsider some things such as the amount of material and resizing the structure to be able to fit a 3 gallon box under the structure. We plan to address this problem by putting the water basin on the floor while the actual structure of the hydroponics sits on the table. Which leads us to our second possible difficulty that might occur when trying to build our prototype would be the water from the basin not pumping properly up to the structure where the plants are planted. Since the design that we plan to build is on a CAD software and it has the water basin located on the floor and the actual hydroponics structure is on the table that our stakeholder wants it to be placed on. We are worried that the water pump would not be powerful enough to pump the water from a low point to a high point. This of course would alter the price of the whole project whether it is requiring more material to build, increasing the price/budget or requiring less material to build it, decreasing the price/budget and it may also take extra time and thinking to find a better solution/better place to put the water basin. Our plans to address this obstacle is possibly putting the water basin next to the hydroponics and making the PVC pipes detachable/disassemble so it can accommodate our stakeholders criteria of it being easy to put into storage. Another difficulty may be if our design can accommodate the type of plants our stakeholder wants to grow. She wants to grow plants such as strawberries, tomatoes, and bell peppers, which can grow up to between 1 foot to 10 feet tall. Our current design had not taken into account the size and height of the plants being growing, which is an issue. We plan to address this obstacle by installing removal trellises so that our stakeholder can use the trellises at her own discretion. This allows our stakeholders to plant a wider variety of plants and does not just limit them to plants that meet the height of the trellis or if they do not, plants that require trellis like small herbs then they can remove the trellis. Generally speaking the removable trellises would satisfy our user's needs since it gives them more freedom to plant a variety of plants. Budget is a potential issue. Cedar planks are the material we want to use for the wall portions of the structure, but the price of the planks may be over our budget. With this in mind, we need to come up with alternative materials that fit our budget but still have the same results we wanted from the cedar planks. Another obstacle is time. The plan is to have a working design ready for the stakeholders by the end of the school year. This potentially may not happen. To address this we could put in extra hours during the school day, working during a free period, rather than just doing work during our class period. We could also take work home and get things done there and bring it back ready to move ahead. As a solution to the budget issue. We have decided instead of using

Cedar Wood which is slightly over our budget, we would use an alternative wood called treated Pine Wood for the structure of our hydroponics unit since it has similar resistance to decay, insects, and other deteriorating factors as the cedar wood and it being within our budget it is a great alternative for Cedar Wood. Additionally, the treated pine wood is lighter than cedar wood so it would accommodate the users' needs. With these adjustments in mind. The user of the hydroponics would have an end result of a waterproof hydroponic unit that looks aesthetically pleasing. Also they will know they will not have to worry about the damages or problems that would occur with a non-waterproof material. Some obstacles that we may encounter in this process are measurements being incorrect, features of the unit not being correctly shown in the design, additions to design that change other aspects, and unexpected changes that we may have to make to the design. With that in mind our team may have to alter the design because of feedback from some of our stakeholders, which means the team would have to take their feedback and implement it into the 3D model. Our team should be able to address these problems further by doing group evaluations and collaborations to see how we as a team can solve the problem. The team have confirmed that this tactic works with many group sessions having been conducted by both teams to achieve the goal of an exceptional 3D design of the hydroponic unit. In order to ensure an up to standard design the team will need to collaborate as a group on problems, which will address any type of obstacle that comes our way.